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A Sector Tilt That Passed Every In-Sample Test — and Failed Out of Sample

Macro liquidity, equity duration, and the boundary of the QE era

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Executive summary

I built a macro-driven sector tilt for the US market, screened it hard enough to kill two of its three components, and ended up with something that looked good: 20-day Spearman IC of **0.189**, a Newey-West t of **3.14** that holds up under a full year of HAC lags, positive information ratio against a neutral benchmark, and the same sign across **8 of 8** sector pairs.

Then I pushed it back to **2000–2007**, a period that never touched a single design decision. The result:

	2008–2019 (design era)	2000–2007 (out-of-sample)
Median IC (20d)	+0.143	+0.026
Pairs with positive IC	6 / 8	5 / 8
Median net Sharpe	+0.346	−0.171
Pairs with positive Sharpe	8 / 8	0 / 8

The model does not generalise. Not marginally — every single pair flips negative. And the failure has a specific, identifiable mechanism, not a statistical shrug.

The break sits exactly at 2008. The same four-signal model is three to five times stronger after quantitative easing begins than before it.

The conclusion this project actually supports is not “here is a signal.” It is: *the relationship “easier central-bank liquidity → outperformance of long-duration equity” is conditional on central-bank balance-sheet policy being the dominant driver of asset prices. That condition held from 2008; it did not hold before.* Any application of this model has to start by asking whether the condition still holds today.

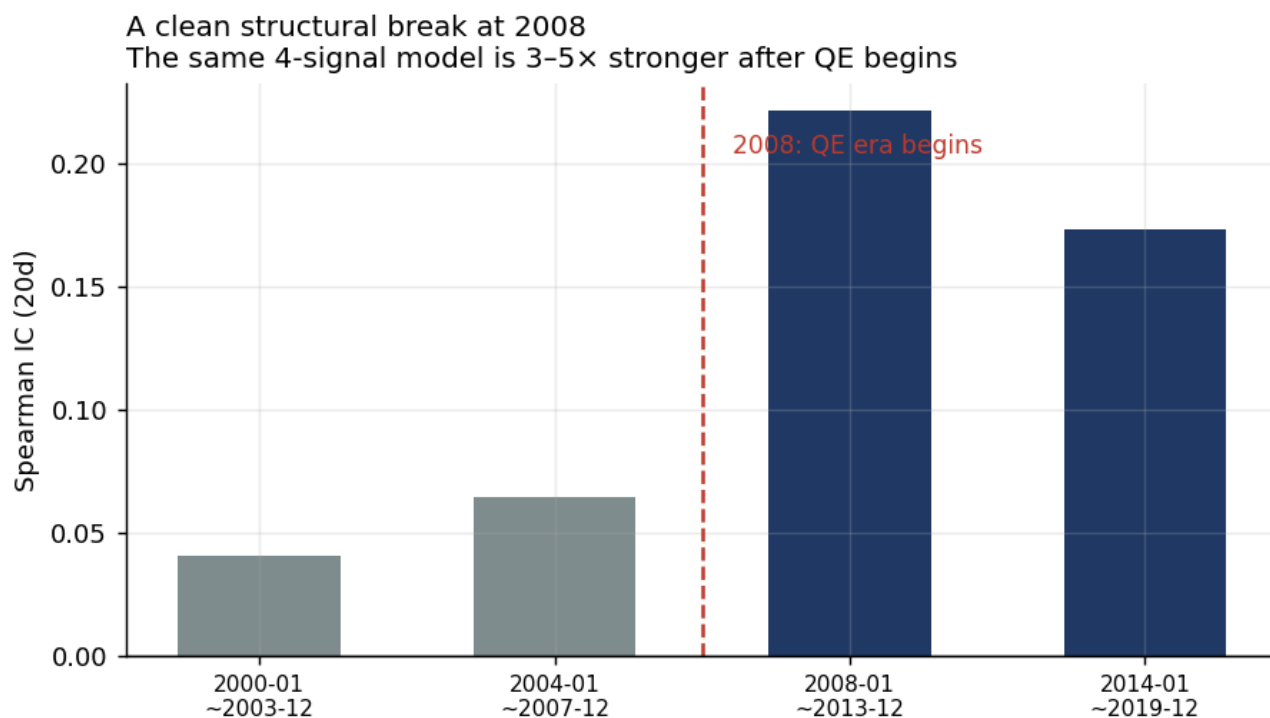


Figure 1: Structural break

1. The framework, briefly

Universe. Relative tilts between US sector ETFs — the decision a sector allocator actually faces (“overweight which of these two?”), not absolute market timing. Common beta cancels out of the spread, which raises signal-to-noise.

Three candidate dimensions, each with signals whose direction priors carry a written economic rationale: momentum & positioning, macro liquidity & risk appetite, relative valuation.

Timing convention. Signal at close $T \rightarrow$ execute at close $T+1 \rightarrow$ returns accrue from $T+2$. Macro series lagged by real publication delay. This convention is enforced identically in IC computation, weight estimation, and backtest.

2. In-sample screening: two of three dimensions rejected

Dimension	IC 20d	Standalone Sharpe	Turnover	Max DD	Verdict
Momentum & positioning	−0.049	−0.126	6.20	−34.2%	Rejected
Macro liquidity	+0.187	+0.390	1.81	−6.8%	Kept
Relative valuation	−0.028	−0.140	3.26	−34.1%	Rejected

Momentum was actively destructive: 3.4× the turnover of the macro dimension, 5× the drawdown, negative return. Equal-weighting all three diluted +0.187 down to **−0.045** and dragged the Newey-West t from +3.14 to −0.59.

A citation I got wrong. I justified the momentum dimension with Moskowitz & Grinblatt (1999). That was a misapplication: their result is *cross-sectional* momentum across ~20 industries,

Two of three dimensions were rejected by their own diagnostics

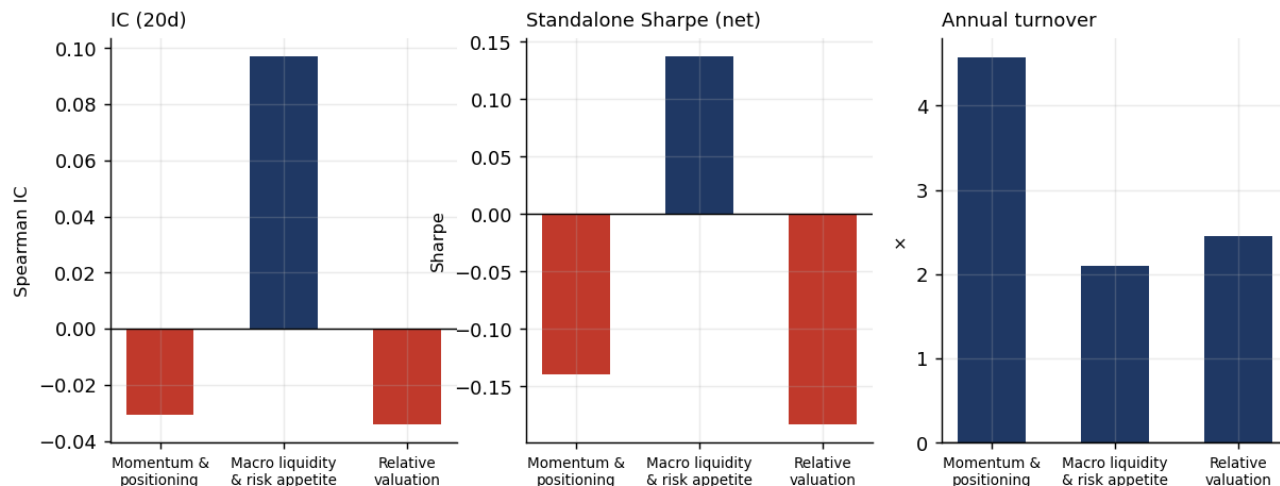


Figure 2: Dimension screening

whereas this is *pairwise* relative momentum between two correlated sectors, where short-horizon mean reversion dominates. An effect documented in one cross-sectional setting does not automatically survive translation to a two-asset pair — a check that should have happened before the signal was built.

The surviving dimension is not one variable in disguise. The equal-weighted composite (IC 0.187) beats every individual signal (best: 0.120), so diversification is genuinely reducing idiosyncratic noise. My prior that the real rate would dominate was also wrong — it is the weakest of the five. The work is done by the Fed balance sheet and the credit spread.

3. In-sample robustness: everything passes

Sub-samples. IC positive in all three sub-periods (0.156 / 0.334 / 0.114), *including* the one where tech actually underperformed — ruling out a disguised buy-and-hold.

Significance under autocorrelation. Slow signals with overlapping windows inflate naive t -statistics, so the full HAC decay curve is reported rather than a single number:

20-day: $t_{\text{naive}} = 11.83 \rightarrow L=22: 3.15 \rightarrow L=63: 2.90 \rightarrow L=126: 3.04 \rightarrow L=252: 3.14$

Extending lags to a full trading year leaves t at 3.14. Signal persistence $\rho_1 = 0.997$, half-life ≈ 215 trading days, 86 sign changes over 18.4 years (≈ 4.7 independent bets per year).

Payoff shape. Asymmetric — nearly all value sits in the top quintile (+1.52%, 70.6% positive); the bottom three are flat and indistinguishable.

Weighting. Equal, inverse-volatility and shrunk IC-weighting are indistinguishable (net Sharpe 0.425 / 0.419 / 0.358) — equal wins, consistent with the general difficulty of beating it out of sample.

Cross-pair. 8 sector pairs, 7/8 positive IC and 8/8 positive Sharpe. And the strength varies exactly as theory predicts: IC correlates +0.85 with the pair's gap in foreign-revenue exposure and +0.70 with the gap in equity duration, but **-0.09 with the gap in beta**. The three near-zero pairs are precisely those without a duration or foreign-revenue gap. That is mechanism validation, and it reframes the signal: this is a *duration and dollar-exposure factor*, not a general rotation tool.

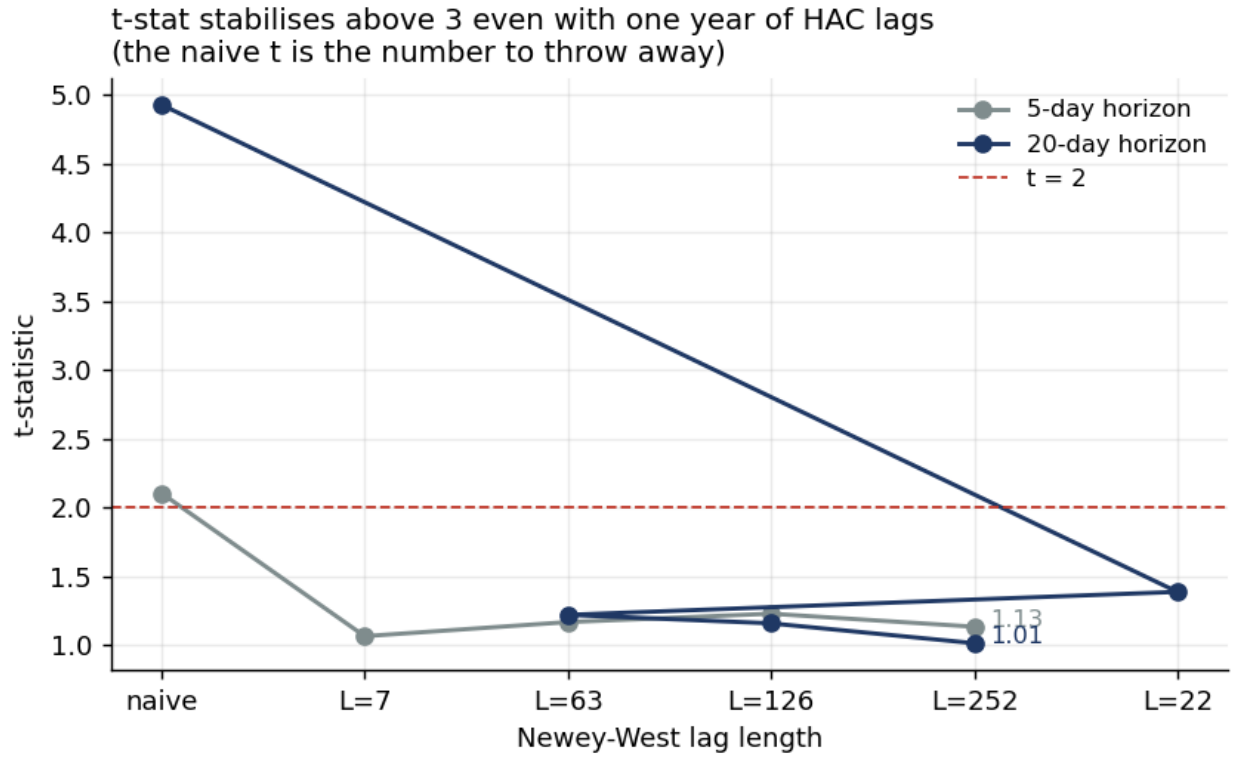


Figure 3: Newey-West decay

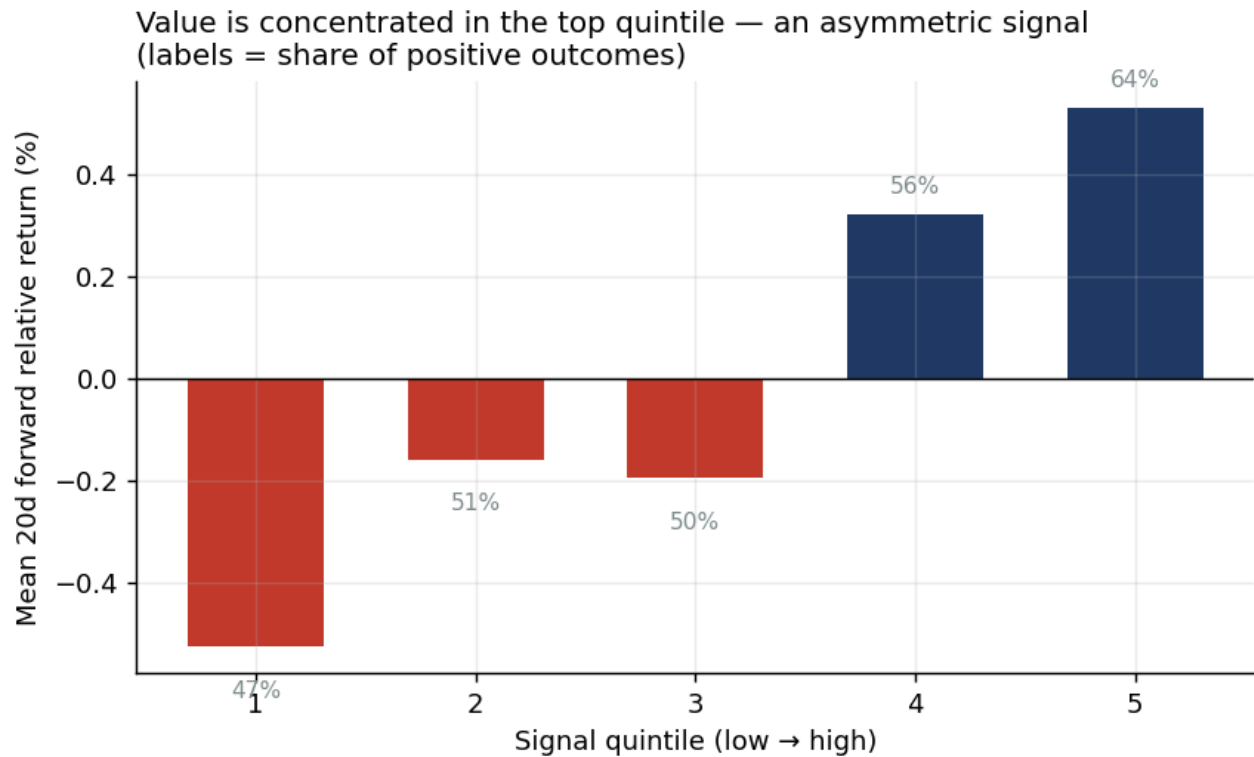


Figure 4: Quintiles

A caveat on that cross-pair evidence. The 8 pairs are not 8 independent tests. Their spread signals have a median pairwise correlation of 0.82, the first principal component explains 77% of variance, and the effective number of independent pairs is roughly **1.6**. Five pairs share tech as the long leg. Pooling their 226 episodes into one sample would have been the single most misleading thing I could have done — the 226 episodes collapse into 54 distinct macro waves.

4. Regime switching: rejected in-sample, vindicated out-of-sample

Fixed direction priors assume the macro-to-sector transmission is invariant. To test that, I pre-declared a set of risk-off multipliers from economic reasoning — never fitted, because ~20 risk-off episodes cannot support estimating five coefficients — and checked whether they helped.

Variant	IC 20d	Net Sharpe	Max DD
Baseline (kept)	0.189	0.425	−6.6%
Regime-switched priors	0.181	0.300	−8.3%
Regime-scaled exposure (0.5×)	0.189	0.408	− 3.8%

Rejected — both IC and Sharpe deteriorated, against a pre-declared acceptance rule. Three of five priors were directionally right (risk signals do bite harder under stress); two were backwards. Most notably, I assumed balance-sheet expansion during stress is “reactive and therefore contaminated” and should be damped. Post-2008 data said the opposite: its IC in risk-off is *double* its IC in risk-on. The 2009 and 2020 QE announcements were the turning points.

Section 5 reverses this verdict. The damping intuition was correct — for the pre-QE era. It failed the post-2008 test only because that sample is dominated by episodes where QE marked the bottom. In 2001–2003 the same reasoning would have saved the model. What looked like a wrong prior was a prior that is *itself* regime-dependent — which is a stronger statement about the model’s instability than the original rejection was.

I did not recalibrate the multipliers to the observed ICs. That would have been exactly the overfitting the pre-declaration existed to prevent.

5. Out-of-sample: 2000–2007

5.1 Design

The 2000–2007 window never informed any design choice — not the dimension screening, not the signal set, not the monthly rebalance, not the hysteresis band.

Three of five macro signals do not exist before 2007 (TIPS real rate from 2003, broad dollar index from 2006, Fed balance sheet from 2002), so long-history substitutes were used: nominal 10-year Treasury for the real rate, the major-currencies dollar index for the broad one, and **the Fed balance sheet was dropped entirely** — no honest long-history substitute exists.

The control that makes this test valid: both periods run the *same* four-signal reduced model. Otherwise “worse out of sample” would be indistinguishable from “fewer signals.” Since the major-currencies dollar index was discontinued in 2020, the comparison window is 2008–2019.

5.2 Result

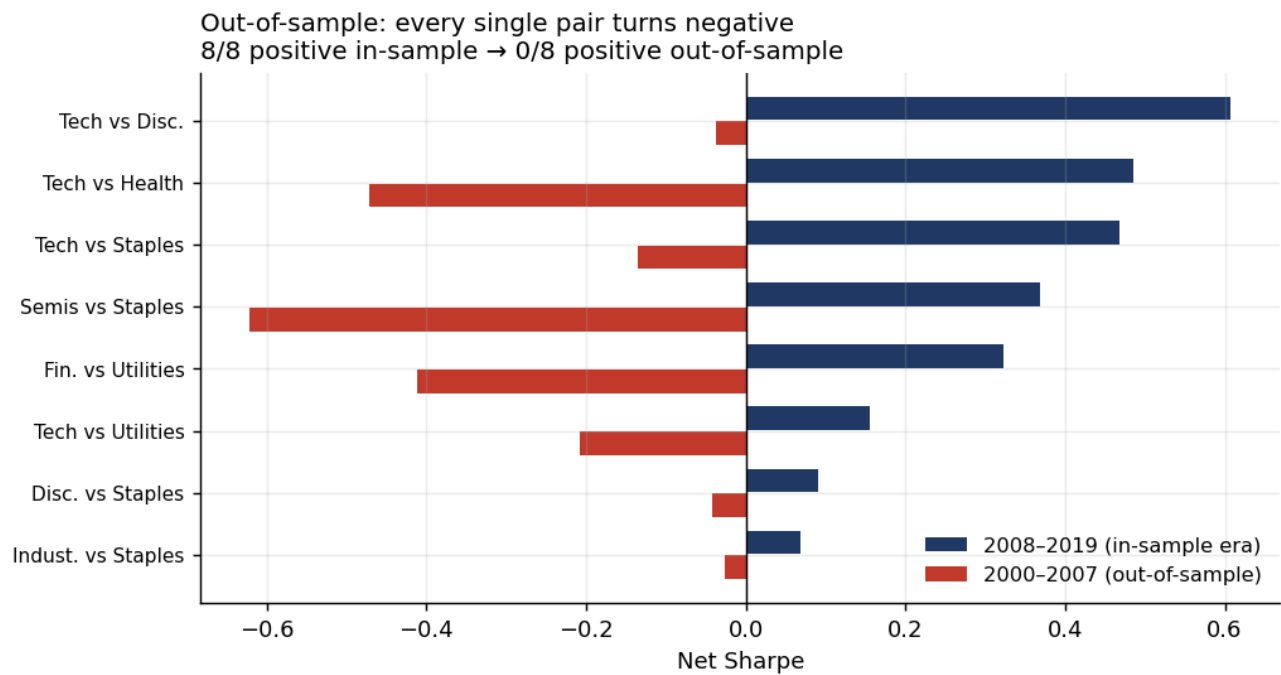
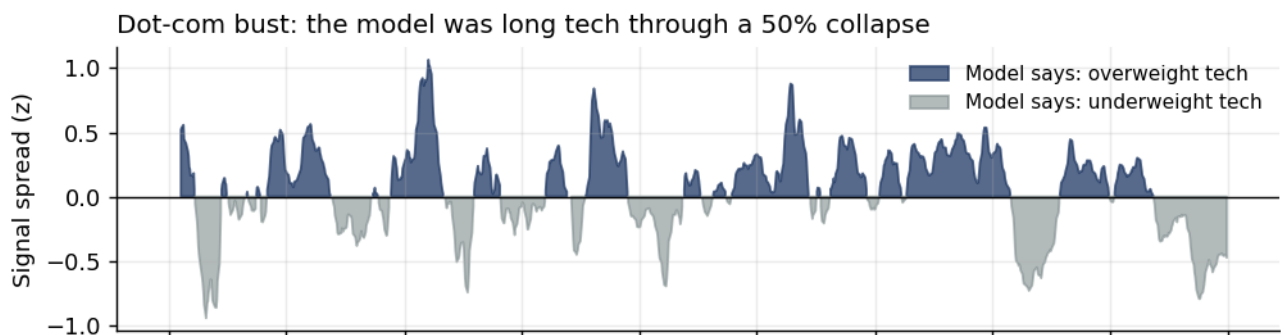


Figure 5: Out-of-sample by pair

	2008-2019	2000-2007
Median IC (20d)	+0.143	+0.026
Median net Sharpe	+0.346	-0.171
Pairs with positive Sharpe	8 / 8	0 / 8

The reduced model still works in 2008-2019 — so the failure is not caused by dropping the Fed balance sheet. The same four signals simply carry no exploitable information before 2008.

5.3 The failure mechanism



2000–2003: +0.041	2004–2007: +0.064
2008–2013: +0.221	2014–2019: +0.173

5.4 What this means

The mechanism requires an environment in which central-bank liquidity is the *dominant marginal driver* of relative equity valuations. That describes 2008–2021 well: balance-sheet policy was the primary tool, and QE announcements repeatedly marked inflection points for growth equities. It does not describe 2000–2007, when rate cuts were reactive to an equity bust and the balance sheet was not a policy instrument at all.

This is a boundary condition, not a bug. But it is a boundary condition wide enough to invalidate the model outside it, and the honest framing is that the in-sample result describes a regime rather than a general market mechanism.

6. What I would actually claim

Narrowly, and with the conditions attached:

Between 2008 and 2026, in US sector pairs with a material gap in equity duration and foreign-revenue exposure, a diversified macro-liquidity composite carried information about one-month relative returns that survived HAC correction, sub-sample splits and weighting-scheme sensitivity. That information was concentrated in stress periods and on the overweight side, and it **does not appear in 2000–2007**, which suggests it is contingent on central-bank balance-sheet policy being the dominant driver of asset prices.

What I would *not* claim: that this is a deployable strategy. Even in-sample it loses to simply holding the tech leg (excess Sharpe -0.25), its excess return is concentrated in 12 of 236 months, and it generates roughly 4.7 independent bets per year. Out-of-sample it loses money in every pair tested.

How it could still be used: as a regime-contingent overweight-confirmation input, subject to an explicit check that the enabling condition still holds — for instance, that central-bank balance-sheet changes still move relative sector valuations. The moment that check fails, the model should be switched off rather than trusted.

7. Method: making the result falsifiable

Thirteen automated anti-look-ahead tests. The strongest is **truncation invariance**: truncate the data at any historical date, re-run the entire signal chain, and every value before the cut must match the full-sample run point for point. Any leakage fails it. Others cover rolling z-score and percentile causality, macro publication lags, forward-return alignment, regime-classification causality, IC-weight realisation lag, a shuffled-signal null, and a perfect-foresight control that must be highly profitable or the engine is wired backwards.

Data. Prices from Tiingo (split- and dividend-adjusted, 1998–present), with Stooq and yfinance as fallbacks and no cross-source mixing within a run. Macro from FRED via the official API.

One data incident worth recording. The credit spread was originally ICE BofA high-yield OAS, which returned only 787 rows starting 2023-08. I diagnosed silent endpoint truncation and wrote multi-path retry logic. The real cause was in the FRED page notes — “*Starting in April 2026, this series will only include 3 years of observations*” — an ICE Data licensing change that no amount of retrying could fix. Substituting Moody’s Baa minus 10-year Treasury, which measures the same premium with 40 years of unrestricted history, raised the most recent sub-period IC from 0.071

to 0.104. Check the data source's own documentation before writing code against an assumed failure mode.

Everything rejected is reproducible. `./run.sh alldims` restores all three dimensions; `--regime` restores regime switching; `./run.sh oos` reruns the out-of-sample test.

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Research exercise. Not investment advice. Backtested results are hypothetical and do not represent actual trading.